TEACHER GUIDE

Pearson Edexcel International GCSE in Business (4BS1)

Business calculations: ARR

Quantitative skills
Calculation: average rate of return



A revision lesson. Pupils run a car dealership in firms of four and revise business calculations: ARR on their own deals. Seat each firm together. Each firm answers the topic tasks together.

The lesson

45-minute setting

Setup	Round 1	Round 2	Round 3	Final questions

EVERY ROUND

Firm decisions
3 min

→

Deals
4½–5 min

→

Topic tasks
4 min

→

B2B deal
2½ min
from Round 2

LESSON SETTINGS

30 min	45 min	60 min	75 min

What pupils do

Every pupil buys and sells cars. In Firm decisions, each role has its own job.



CEO

Staff · Sabotage from Round 2



Finance

Capital & Loans · Investments



Sales & Marketing

Marketing



Operations

Seasonal stock

Topic tasks

Each firm gets its own figures. The method is the same. Your Live screen shows each task and every firm's answer.

1 Lifetime profit

Round 1

A new valeting bay.

Total profit over its life?

Cost to set up	**£45,000**
Returns, each year	**£14,500**
It lasts	**5 years**

ANSWER

£27,500

5 × £14,500 − £45,000 = £27,500

Why: It earns back its cost and £27,500 more.

2 What ARR misses

Round 1

You compare two investments by ARR.

Which two things does ARR not show?

- **A** The cost of the investment
- **B** When the money comes back
- **C** The average profit a year
- **D** The risk that returns fall short
- **E** Profit as a % of the cost

ANSWER

✓ When the money comes back
✓ The risk that returns fall short

Why: ARR averages the profit; it says nothing of timing or risk.

3 ARR steps in order

Round 2

You work out the ARR of a new ramp.

In what order do you work it out?

- **A** Divide by the years
- **B** Add up all the returns
- **C** Divide by the cost, × 100
- **D** Take off the cost: total profit

ANSWER

1. Add up all the returns
2. Take off the cost: total profit
3. Divide by the years
4. Divide by the cost, × 100

Why: Total profit, then profit a year, then as a % of the cost.

Topic tasks

4 Valeting bay ARR

Round 2

A new valeting bay.

Average rate of return (ARR), as a %?

Cost to set up	**£60,000**
Profit over its life	**£84,000**
It lasts	**5 years**

ANSWER

28%

£84,000 ÷ 5 = £16,800; ÷ £60,000 × 100 = 28%

Why: It earns 28p a year for every £1 put in.

5 ARR below loan rate

Round 3

A ramp would have an ARR of 6%. You would borrow the money at 8% a year.

Should you invest, and why?

A No

B Yes

ANSWER

✔ **No**

Why: A 6% return does not cover an 8% cost of borrowing.

6 Better option ARR

Round 3

Which one pays better?

ARR of the better option, as a %?

Option A: cost	**£75,000**
Option A: profit each year	**£9,000**
Option B: cost	**£55,000**
Option B: profit each year	**£13,750**

ANSWER

25%

A 12%; B 25%; better 25%

Why: Compare the rates, not the pounds of profit.

Final questions

Each pupil answers 6 final questions: general International GCSE business questions.

Then each pupil writes one short answer: “Explain why ARR is compared with the interest rate.”